# Mock Paper 1 — Computer Systems (H446/01)

Time allowed: 2 hours 30 minutes · Total: 140 marks · Answer all questions.

Instructions: Show all working on calculations and logic questions. Mark against the mark scheme at the end.

AO key: **AO1** = knowledge & understanding · **AO2** = application · **AO3** = analysis/evaluation/design.

Running mark tally is shown beside each question. Whole-paper total is verified at the very end.

## Section A / questions

### Question 1 — Processor architecture and the FDE cycle *(8 marks)*

A CPU uses the Von Neumann architecture and contains the registers PC, MAR, MDR, CIR and ACC.

**1(a)** State the role of the **Program Counter (PC)** and the **Memory Address Register (MAR)** during the fetch stage. **[3]** *(AO1)*

**1(b)** Describe **three** factors, other than clock speed, that affect the performance of a processor. **[3]** *(AO1)*

**1(c)** Explain **one** advantage of a Harvard architecture over a Von Neumann architecture for an embedded system. **[2]** *(AO2)*

*(Running total: 8)*

### Question 2 — Storage devices *(7 marks)*

**2(a)** State **two** characteristics of a solid-state drive (SSD) that make it suitable for use in a laptop. **[2]** *(AO1)*

**2(b)** Compare optical storage with magnetic storage, giving **three** distinct points of comparison. **[3]** *(AO2)*

**2(c)** A digital camera stores photographs on a removable memory card. State the **type** of storage technology used and justify why it is appropriate. **[2]** *(AO2)*

*(Running total: 15)*

### Question 3 — Operating systems *(9 marks)*

**3(a)** Explain how a **paging** memory-management technique allows a program larger than physical RAM to run. **[4]** *(AO1)*

**3(b)** Describe the purpose of an **interrupt** and explain the role of the **interrupt service routine (ISR)**. **[3]** *(AO1)*

**3(c)** State **one** difference between a **multi-tasking** and a **multi-user** operating system. **[2]** *(AO1)*

*(Running total: 24)*

## Question 4 — Translators and the compilation process *(9 marks)*

**4(a)** State **three** differences between a **compiler** and an **interpreter**. **[3]** *(AO1)*

**4(b)** The compilation of source code passes through four stages: lexical analysis, syntax analysis, code generation and optimisation. Describe what happens during **lexical analysis** and **syntax analysis**. **[4]** *(AO1)*

**4(c)** Explain why a programmer might choose to distribute software as **bytecode** for a virtual machine rather than as native machine code. **[2]** *(AO2)*

*(Running total: 33)*

## Question 5 — Software development methodologies *(7 marks)*

**5(a)** A start-up is building a mobile app where the client's requirements are expected to change frequently. Justify why an **Agile** methodology would be more suitable than the **Waterfall** model. **[4]** *(AO2)*

**5(b)** Describe **three** features of the **Extreme Programming (XP)** methodology. **[3]** *(AO1)*

*(Running total: 40)*

## Question 6 — Object-oriented programming *(9 marks)*

A program models library members. A base class `Member` has the attributes `name` and `memberID`. A subclass `StudentMember` adds the attribute `course` and overrides the method `borrowLimit()`.

**6(a)** Explain what is meant by **inheritance** and **encapsulation**, using this scenario to illustrate each. **[3]** *(AO2)*

**6(b)** Write the class definition for `Member` in pseudocode (or a language of your choice), including a constructor and a `getDetails()` method. Use private attributes with public accessor methods. **[4]** *(AO3)*

**6(c)** Explain what is meant by **polymorphism** and give an example using `borrowLimit()`. **[2]** *(AO2)*

*(Running total: 49)*

## Question 7 — Assembly language and addressing modes *(8 marks)*

A processor uses the Little Man Computer (LMC)-style instruction set. Consider this program:

```
        INP
        STA  num
loop    LDA  total
        ADD  num
        STA  total
        LDA  num
        SUB  one
        STA  num
        BRP  loop
        LDA  total
        OUT
        HLT
num     DAT
total   DAT  0
one     DAT  1
```

**7(a)** Trace the program for an input of **3**. State the final value output and explain in one sentence what the program computes. **[4]** *(AO3)*

**7(b)** Explain the difference between **immediate** and **direct (absolute)** addressing, and state **one** reason a processor also provides **indexed** addressing. **[4]** *(AO1)*

*(Running total: 57)*

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## Question 8 — Compression, encryption and hashing *(10 marks)*

**8(a)** Explain the difference between **lossy** and **lossless** compression, giving **one** suitable use case for each. **[3]** *(AO1)*

**8(b)** Describe how **run-length encoding (RLE)** compresses data and state **one** type of data for which it is ineffective. **[3]** *(AO2)*

**8(c)** Explain how **symmetric** encryption differs from **asymmetric** encryption, and describe how asymmetric encryption can be used to create a **digital signature**. **[4]** *(AO2)*

*(Running total: 67)*

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## Question 9 — Databases and SQL *(10 marks)*

A `Booking` table and a `Customer` table store data for a car-hire company:

```
Customer(CustomerID, Surname, City)
Booking(BookingID, CustomerID, CarReg, StartDate, Days)
```

**9(a)** State what is meant by a **primary key** and a **foreign key**, identifying one of each from the tables above. **[2]** *(AO1)*

**9(b)** Write an SQL statement to list the `Surname` and `City` of every customer who lives in `'Leeds'`, sorted alphabetically by `Surname`. **[3]** *(AO3)*

**9(c)** Write an SQL statement that lists the `Surname` of each customer together with the `CarReg` and `StartDate` of all their bookings of **5 or more** days. **[5]** *(AO3)*

*(Running total: 77)*

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## Question 10 — Networks and TCP/IP *(10 marks)*

**10(a)** Describe the role of each of the four layers of the **TCP/IP stack**. **[3]** *(AO1)*

**10(b)** Explain how the **Domain Name System (DNS)** resolves a URL into an IP address, referring to the hierarchy of DNS servers. **[4]** *(AO2)*

**10(c)** Explain the difference between **circuit switching** and **packet switching**, and state why packet switching is used on the internet. **[3]** *(AO2)*

*(Running total: 87)*

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## Question 11 — Number representation *(12 marks)*

**11(a)** Convert the denary number **−42** into **8-bit two's complement**. Show your working. **[3]** *(AO2)*

**11(b)** A floating-point number uses an **8-bit mantissa** and a **4-bit exponent**, both in two's complement, with an assumed binary point after the first mantissa bit. The stored value is:

```
mantissa = 0101 1000   exponent = 0011
```

Calculate the denary value this represents. Show your working, and state whether the number is **normalised**, justifying your answer. **[5]** *(AO3)*

**11(c)** Perform the binary addition `0110 1101 + 0101 0011` in 8 bits. State the 8-bit result and explain whether **overflow** has occurred if the values are interpreted as signed two's complement. **[4]** *(AO3)*

*(Running total: 99)*

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## Question 12 — Data structures *(10 marks)*

**12(a)** Explain the difference between a **stack** and a **queue**, and give **one** real use of each in a computer system. **[3]** *(AO1)*

**12(b)** A **circular queue** is implemented in an array of size 5 with `front` and `rear` pointers. Explain **why** a circular queue is preferred to a linear queue, and describe how the `rear` pointer wraps around. **[4]** *(AO2)*

**12(c)** State the **Big O** time complexity of: (i) adding an item to the top of a stack; (ii) a linear search of an unsorted list of *n* items; (iii) a binary search of a sorted list of *n* items. **[3]** *(AO1)*

*(Running total: 109)*

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## Question 13 — Boolean algebra and logic circuits *(12 marks)*

**13(a)** Complete the truth table for the expression `Q = (A AND NOT B) OR (NOT A AND B)`, and state the name of the single logic gate that this expression is equivalent to. **[4]** *(AO3)*

**13(b)** Using the laws of Boolean algebra, simplify the expression `Q = A.B + A.(B + C) + B.(B + C)` to its simplest form. Show each step and name the law used. **[4]** *(AO3)*

**13(c)** A safety system sounds an alarm (`Z = 1`) when **either** a door sensor `D` is open **and** the system is armed `A`, **or** when a panic button `P` is pressed (regardless of the other inputs). Write the Boolean expression for `Z` and draw the logic circuit using AND, OR and any other required gates. **[4]** *(AO3)*

*(Running total: 121)*

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## Question 14 — Networking and security *(7 marks)*

**14(a)** Explain the difference between a **LAN** and a **WAN**, referring to ownership and geographical scale. **[2]** *(AO1)*

**14(b)** Describe how a **firewall** using **packet filtering** helps protect a network. **[3]** *(AO2)*

**14(c)** Explain why a **star** network topology is more robust than a **bus** topology when a single connection fails. **[2]** *(AO2)*

*(Running total: 128)*

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## Question 15 — Extended response: legal, moral and ethical issues *(12 marks)*

A large retailer introduces an **AI-driven facial-recognition system** in its stores. The system scans the faces of every customer who enters, matches them against a database of known shoplifters, and alerts security staff in real time. Camera footage and biometric data are stored on the company's servers for 12 months.

**15** Discuss the **legal, ethical and privacy issues** raised by this system. In your answer you should consider the interests of customers, the retailer and society, and reach a justified conclusion about whether the system should be deployed. **[12]** *(AO3)*

*(Running total: 140)*

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## Mark scheme

> General guidance: award marks for any valid equivalent point. Where a question says "three differences", award one mark per distinct correct point up to the maximum. Calculation questions: award method marks even if the final answer is wrong, provided working is shown.

### Question 1 (8 marks)

**1(a) [3] (AO1)** — 1 mark each, max 3: - PC holds the **address of the next instruction** to be fetched. - At the start of fetch this address is **copied from the PC into the MAR**. - The MAR holds the **address of the memory location** to be read (or written); the address bus carries this address to memory. (PC is then incremented.)

**1(b) [3] (AO1)** — 1 mark each, max 3 (any three): - **Number of cores** — more cores allow more instructions/processes to be executed in parallel. - **Cache size/levels** — a larger cache reduces the need to fetch from slower main memory. - **Word length / bus width** — wider data/address buses move more data per cycle / address more memory. - **Pipelining** — fetching, decoding and executing different instructions simultaneously.

**1(c) [2] (AO2)** — 1 mark for advantage stated, 1 for explanation in context: - Harvard has **separate memories/buses for instructions and data**, so an instruction and a data item can be fetched **simultaneously** (1), increasing throughput — useful in an embedded/real-time system where speed and predictable timing matter (1). (Accept: instruction and data memories can have different widths/optimisations.)

### Question 2 (7 marks)

**2(a) [2] (AO1)** — 1 each, max 2: no moving parts so more durable/shock-resistant; lower power consumption (longer battery life); faster access/read-write times; silent; lighter.

**2(b) [3] (AO2)** — 1 mark per valid comparison point, max 3: - Magnetic typically offers **higher capacity per £** than optical. - Optical media (e.g. Blu-ray) are **cheap to mass-produce/distribute**; magnetic (HDD) is re-writable many times more reliably. - Magnetic generally has **faster access/transfer** than optical drives. - Optical is more portable/robust against magnetic fields; magnetic can be erased by strong magnetic fields.

**2(c) [2] (AO2)** — 1 + 1: - **Solid state / flash memory** (1). - Justification: no moving parts so robust for a portable device, low power, fast write, small and removable (any one valid) (1).

### Question 3 (9 marks)

**3(a) [4] (AO1)** — 1 mark each, max 4: - Memory (physical and virtual) is divided into fixed-size blocks called **pages/frames**. - Pages not currently needed are stored on **secondary storage (the swap/page file)**. - When a program needs a page that is not in RAM (a **page fault**), the OS **swaps** a page out and loads the required page in. - This allows the **logical address space to exceed physical RAM** / programs larger than RAM to run (accept: addresses are translated logical→physical via a page table).

**3(b) [3] (AO1)** — max 3: - An interrupt is a **signal to the processor** that an event needs attention (from hardware/software) (1). - The current state/registers are **saved (pushed to a stack)** before servicing (1). - The **ISR is the routine that handles that specific interrupt**; afterwards the saved state is restored and execution resumes (1).

**3(c) [2] (AO1)** — 2 marks for a clear distinction: - **Multi-tasking**: one user appears to run several processes "at once" via time-slicing/scheduling. - **Multi-user**: several users share the resources of one system simultaneously (the OS allocates processor time/resources between them and keeps their data separate).

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## Question 4 (9 marks)

**4(a) [3] (AO1)** — 1 per difference, max 3: - Compiler translates the **whole program at once** to produce an executable; interpreter translates and executes **line by line**. - Compiled code **runs faster**; interpreted code is generally slower (re-translated each run). - Compiler reports **all errors after compilation**; interpreter stops at the **first error** encountered. - Compiled executable is **platform-specific** and can be distributed without source; interpreter needs to be present at run time.

**4(b) [4] (AO1)** — 2 + 2: - **Lexical analysis (2):** source code is broken into **tokens**; whitespace and comments are removed; identifiers are entered into the **symbol table**. - **Syntax analysis (2):** tokens are checked against the **grammar/rules of the language** (often building a parse/abstract syntax tree); **syntax errors are reported** if the structure is invalid.

**4(c) [2] (AO2)** — 1 + 1: - Bytecode is **platform-independent** — the same file runs on any device with the appropriate **virtual machine** (1). - So the developer writes/distributes **once** rather than recompiling for each platform / source code is not exposed (1).

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## Question 5 (7 marks)

**5(a) [4] (AO2)** — up to 4 for justified application: - Agile is **iterative**, delivering working increments, so changing requirements can be incorporated each iteration (1) whereas Waterfall fixes requirements up front (1). - **Frequent client involvement/feedback** ensures the app matches evolving needs (1). - Waterfall would require costly return to earlier stages if requirements change late (1); Agile reduces the risk of building the wrong product (1). (Max 4.)

**5(b) [3] (AO1)** — 1 each, max 3: pair programming; test-driven development / writing tests first; frequent small releases; continuous integration; on-site customer; short iterations/stand-ups; collective code ownership; refactoring.

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## Question 6 (9 marks)

**6(a) [3] (AO2)** — inheritance up to 2, encapsulation up to 1 (or 2+1 split, max 3): - **Inheritance**: `StudentMember` inherits the attributes/methods of `Member` (e.g. `name`, `memberID`) and adds/overrides its own (`course`, `borrowLimit()`) — reuse without rewriting (1–2). - **Encapsulation**: attributes such as `memberID` are kept **private** and accessed only through public methods, protecting the data from invalid external changes (1).

**6(b) [4] (AO3)** — award: - Private attributes declared (1) - Constructor that initialises both attributes (1) - Public getter/accessor method(s) (1) - `getDetails()` returns/combines the details correctly (1)

Example (accept any valid OOP language):

```
class Member
    private name
    private memberID

    public procedure new(name, memberID)
        self.name = name
        self.memberID = memberID
    endprocedure

    public function getName()
        return self.name
    endfunction

    public function getDetails()
        return self.name + " (" + self.memberID + ")"
    endfunction
endclass
```

**6(c) [2] (AO2)** — 1 + 1: - Polymorphism: the **same method name behaves differently depending on the object's class** (1). - Example: calling `borrowLimit()` on a `Member` returns the base limit, but on a `StudentMember` the **overridden** version runs and returns a different limit (1).

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## Question 7 (8 marks)

**7(a) [4] (AO3)** — trace + result + explanation:

Input `num = 3`, `total = 0`, `one = 1`. The loop adds `num` to `total`, then decrements `num`, repeating while `num ≥ 0` (BRP = branch if accumulator positive, i.e. ≥ 0).

| Pass | num at start | total += num | num after SUB one |
|---|---|---|---|
| 1 | 3 | 0+3 = 3 | 2 |
| 2 | 2 | 3+2 = 5 | 1 |
| 3 | 1 | 5+1 = 6 | 0 |
| 4 | 0 | 6+0 = 6 | −1 → BRP false, exit |

- Correct trace of total (1) and of num decrement (1).
- **Output = 6** (1).
- Explanation: it computes the **sum 3+2+1+0 = the sum of all integers from the input down to 0** (i.e. triangular sum) (1).

**7(b) [4] (AO1)** — 1 + 1 + 1 + 1: - **Immediate**: the operand **is the actual value** to be used (no memory lookup) (1). - **Direct/absolute**: the operand is the **address of the data** in memory; the value at that address is used (1). - Distinction clearly drawn (1). - **Indexed**: the effective address = base address + contents of an **index register**, which makes it easy/efficient to **step through arrays/lists** (1).

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### Question 8 (10 marks)

**8(a) [3] (AO1)** — 2 for distinction, 1 for use cases: - **Lossless**: original data can be **perfectly reconstructed**; no data discarded. - **Lossy**: some data is **permanently discarded** to achieve smaller files; original cannot be exactly recovered. - Use cases: lossless for text/program files/ZIP; lossy for streaming audio/video/JPEG images (1 for both).

**8(b) [3] (AO2)** — 2 + 1: - RLE replaces **runs of repeated identical values** with a single value plus a **count** (e.g. `AAAA` → `4A`) (2). - Ineffective for data with **few/no repeats**, e.g. natural photographic images or already-compressed/random data — may even increase size (1).

**8(c) [4] (AO2)** — 2 + 2: - **Symmetric**: the **same single key** encrypts and decrypts; key must be shared securely (1). **Asymmetric**: a **public key** encrypts and a different **private key** decrypts (a key pair) (1). - **Digital signature**: the sender creates a **hash/digest** of the message and encrypts it with **their private key** (1); the recipient decrypts it with the sender's **public key** and compares hashes — confirming the sender's identity and that the message is unaltered (1).

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### Question 9 (10 marks)

**9(a) [2] (AO1)** — 1 + 1: - **Primary key**: an attribute that **uniquely identifies each record** in a table — e.g. `CustomerID` (in Customer) or `BookingID` (in Booking). - **Foreign key**: an attribute that is a primary key **in another table**, used to link tables — e.g. `CustomerID` in `Booking`.

**9(b) [3] (AO3)** — SELECT correct fields (1), correct WHERE (1), correct ORDER BY (1):

```
SELECT Surname, City
FROM Customer
WHERE City = 'Leeds'
ORDER BY Surname;
```

(Accept `ORDER BY Surname ASC`.)

**9(c) [5] (AO3)** — SELECT correct fields (1), both tables in FROM (1), correct join condition (1), correct filter `Days >= 5` (1), valid overall syntax (1):

```
SELECT Customer.Surname, Booking.CarReg, Booking.StartDate
FROM Customer, Booking
WHERE Customer.CustomerID = Booking.CustomerID
AND Booking.Days >= 5;
```

(Accept explicit `INNER JOIN ... ON ...` form.)

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### Question 10 (10 marks)

**10(a) [3] (AO1)** — award up to 3 across the four layers (1 per correct role, max 3): - **Application** layer: provides protocols for applications (HTTP, FTP, SMTP) / interface to the user's program. - **Transport** layer: splits data into **segments/packets**, manages end-to-end connection, ports, and reliability (TCP/UDP). - **Internet (Network)** layer: adds **IP addresses** and routes packets across networks. - **Link (Network access)** layer: handles the **physical/MAC** transmission on the local network.

**10(b) [4] (AO2)** — up to 4: - Browser/resolver first checks a **local cache**; if not found it queries a DNS server (1). - DNS servers are arranged in a **hierarchy**: root servers → **top-level domain (TLD)** servers (e.g. `.com`, `.uk`) → **authoritative** name servers for the domain (1). - The query is passed up/along this hierarchy until the **authoritative server returns the IP address** for the domain (1). - The IP address is returned to the host (and cached) so the browser can connect to the correct server (1).

**10(c) [3] (AO2)** — 1 + 1 + 1: - **Circuit switching** establishes a **dedicated physical path** for the whole communication (e.g. a phone call) (1). - **Packet switching** splits data into packets that are **routed independently** and reassembled at the destination (1). - Packet switching is used on the internet because it **uses bandwidth efficiently / shares links / reroutes around failures** (no dedicated line tied up) (1).

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### Question 11 (12 marks)

**11(a) [3] (AO2)** — method + answer: - +42 in binary = `0010 1010` (1). - Invert: `1101 0101`; add 1 (1). - Result **`1101 0110`** (1). (Check: −128+64+16+4+2 = −42. ✓)

**11(b) [5] (AO3)** — - Exponent `0011` (two's complement) = **+3** (1). - Mantissa `0.1011000` with point after first bit; bit values from the point: 0.1011000 → $1 \times 2^{-1} + 0 + 1 \times 2^{-3} + 1 \times 2^{-4} = 0.5 + 0.125 + 0.0625 =$ **0.6875**, with a leading `0` sign bit so it is **positive** (1 for mantissa value). - Apply exponent: $0.6875 \times 2^3 = 0.6875 \times 8 =$ **5.5** (1 for shifting/multiplying, 1 for final value **5.5**). - **Normalised? Yes** — for a positive number the first two bits are `0` then `1` (`01…`), which is the normalised form; the leading bits differ so no precision is wasted (1).

(Full marks: exponent value, mantissa value, multiplication, final answer 5.5, correct normalised justification.)

**11(c) [4] (AO3)** —

```
  0110 1101   (= 109)
+ 0101 0011   (=  83)
-----------
  1100 0000   (= 192 unsigned)
```

- Correct bit-by-bit addition / carries (1).
- **8-bit result `1100 0000`** (1).
- Interpreted as signed two's complement: `0110 1101` = +109 and `0101 0011` = +83; sum = +192 which **exceeds +127** (1).
- Therefore **overflow has occurred** — two positives gave a negative result (`1100 0000` = −64 in two's complement), i.e. the sign bit changed incorrectly (1).

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## Question 12 (10 marks)

**12(a) [3] (AO1)** — 2 for distinction, 1 for uses: - **Stack**: **LIFO** (last in, first out). **Queue**: **FIFO** (first in, first out) (2 for both, or 1 each). - Uses (1, any): stack — managing subroutine return addresses / undo / call stack; queue — print/job spooling / keyboard buffer / scheduling.

**12(b) [4] (AO2)** — up to 4: - In a **linear** queue, once `rear` reaches the end of the array the space freed at the **front by dequeuing cannot be reused**, wasting memory (1). - A **circular** queue treats the array as a ring so freed front slots can be **reused**, making fuller use of fixed memory (1). - The `rear` pointer wraps using **modulo arithmetic**: `rear = (rear + 1) MOD size` (1). - So after the last index it returns to index 0 (provided the queue is not full), keeping enqueue O(1) (1).

**12(c) [3] (AO1)** — 1 each: - (i) Push to stack = **O(1)**. - (ii) Linear search = **O(n)**. - (iii) Binary search = **O(log n)**.

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## Question 13 (12 marks)

**13(a) [4] (AO3)** — truth table (3) + gate name (1):

| A | B | NOT A | NOT B | A AND NOT B | NOT A AND B | Q |
|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | **0** |
| 0 | 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 0 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 0 | 0 | 0 | **0** |

- 3 marks for a correct Q column (deduct 1 per error, min 0; award 1 if at least half correct).
- The expression is equivalent to an **XOR gate** (1).

**13(b) [4] (AO3)** — award 1 per correct step/law, max 4:

```
Q = A.B + A.(B + C) + B.(B + C)
  = A.B + A.B + A.C + B.B + B.C        (distribution)
  = A.B + A.C + B + B.C                 (A.B + A.B = A.B ; B.B = B)
  = A.B + A.C + B                       (B + B.C = B, absorption)
  = A.C + B                             (B + A.B = B, absorption)
```

- **Simplest form: `Q = B + A.C`** (final answer mark).
- (Accept correct alternative ordering of laws; key laws: distribution, idempotence B.B=B, absorption.)

**13(c) [4] (AO3)** — - Expression: **`Z = (D AND A) OR P`** (2 marks: AND of D,A = 1; OR with P = 1). - Circuit (2 marks: D and A into an AND gate; output of AND and P into an OR gate; output = Z):

```
D ──┐
    │  AND ──┐
A ──┘        │
             OR ── Z
P ───────────┘
```

- 1 mark for correct AND gate on D, A; 1 mark for OR gate feeding from the AND output and P to give Z.

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### Question 14 (7 marks)

**14(a) [2] (AO1)** — 1 + 1: - **LAN**: covers a **small geographical area** (single site/building) and the **infrastructure is owned by the organisation** (1). - **WAN**: covers a **large geographical area** (between sites/countries) and often uses **third-party/leased** communication links (1).

**14(b) [3] (AO2)** — up to 3: - A firewall sits between the internal network and external network and **inspects packets** entering/leaving (1). - **Packet filtering** examines each packet's header — e.g. **source/destination IP address and port number** — against a set of rules (1). - Packets that do not meet the rules are **dropped/blocked**, preventing unauthorised traffic/ports from reaching the network (1).

**14(c) [2] (AO2)** — 1 + 1: - In a **bus** topology all devices share a single backbone, so a break in the cable can bring down the **whole network** (1). - In a **star** topology each device has its own link to a central switch, so if one link fails **only that device** is affected; the rest continue to communicate (1).

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### Question 15 — Extended response (12 marks) (AO3)

**Indicative content** (not exhaustive — credit any relevant, well-argued point across legal, ethical, privacy and stakeholder dimensions):

*Legal:* - Data Protection Act 2018 / GDPR: biometric (facial) data is **special-category personal data** requiring a lawful basis and explicit safeguards; 12-month retention must be justified and proportionate. - Computer Misuse / security obligations to keep the biometric database secure; risk of breach. - Customers' rights: to be informed, to access, and potential lack of meaningful **consent** when scanning is automatic on entry.

*Ethical / moral:* - **Presumption of innocence** — scanning every customer treats all as potential suspects. - **Algorithmic bias / false positives** — facial recognition is known to be less accurate for some demographic groups, risking wrongful accusation and discrimination. - Transparency and accountability for decisions made by the system.

*Privacy:* - Mass surveillance of ordinary customers; tracking of movements; potential "function creep" (data reused for marketing/other purposes). - Storing footage and biometrics for 12 months increases the impact of any breach.

*Stakeholder perspectives:* - **Retailer**: reduced theft/loss, staff safety, cost savings vs reputational and legal risk. - **Customers**: safer stores vs loss of privacy and risk of misidentification. - **Society**: deterrence of crime vs normalising surveillance and eroding civil liberties.

*Conclusion:* a justified judgement (e.g. deploy only with clear signage, minimal retention, human review of all matches, bias testing and a lawful basis — or do not deploy because consent and proportionality cannot be met).

**Levels of response:**

**Level 3 (9–12 marks):** A thorough discussion covering legal, ethical and privacy issues with clear application to the scenario. Considers **multiple stakeholders** and weighs competing interests. Reaches a

**well-justified, balanced conclusion**. Specialist terminology used accurately; response is coherent, structured and sustained.

**Level 2 (5–8 marks):** Discusses several relevant issues with some application to the scenario. Considers more than one viewpoint but with limited balance/depth. A conclusion is offered but **partially justified**. Some structure and reasonable use of terminology.

**Level 1 (1–4 marks):** Identifies a few relevant points, largely descriptive/generic with little application to the scenario. One-sided; little or no justified conclusion. Limited structure.

**0 marks:** No creditable response.

## Verified mark total

8 + 7 + 9 + 9 + 7 + 9 + 8 + 10 + 10 + 10 + 12 + 10 + 12 + 7 + 12 = **140 marks**.

**Confirmed total: 140 marks.**